UW

Department of Computer Science Lab Sessional

Course Code: CS-201L

Course Title: Data Structure Lab

Total Marks: 25

Date: __/__/2025

Name: _____

Registration No.

_____

Discipline/Semester: BSCS-**3**B

Instructor: Muniba Khan

Q1. Perform the following task [LL02, GA4, P4] **Total marks:25**

Scenario:

You are hired as a **C++ Developer** to build a **Student & Inventory Management System**. The system must use arithmetic operations, decision statements, loops, arrays, pointers, structures, dynamic memory, and linked lists.

TASK 1: (5 Marks)

Write C++ code for the following:

1. Declare the variables int quantity, float price, and totalBill for the inventory section.
2. Take input from the user for **quantity** and **price**, compute the total bill using totalBill = quantity * price, and display the result.
3. Using two sample integer values, perform and display **addition, subtraction, multiplication, division, and modulus**.

Write C++ code for the following:

1. Input marks of a student and display the grade using **if-else** or **switch-case**.
2. Use a **for loop** to print numbers from 1 to 10.
3. Use a **while loop** to calculate and display the sum of numbers from 1 to N, where N is entered by the user.

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4. Use a **do-while loop** to display the multiplication table of a number entered by the user.

Task 2 – (5 Marks)

Array Operations & Search

Consider the array below representing stock quantities:

```
int stock[10] = {5, 15, 22, 35, 40, 55, 65, 70, 85, 95};
```

Write C++ code for the following:

1. Implement **linear search** to find a specific quantity.
2. Insert a new quantity at a specific position in the array.
3. Delete an element from a given position in the array.
4. Find and display the **maximum** and **minimum** quantity in the updated array.
5. Display the total size of the array using sizeof().

TASK 3 – (5 Marks)

2D Arrays & Structures

Student	Marks in 3 Subjects
Student1	[80, 85, 78]
Student2	[88, 82, 90]
Student3	[75, 89, 84]

Write a C++ program to:

1. Declare a **2D array** for student marks.
2. Display the marks in **matrix form**.
3. Use pointer variables and the & operator to display addresses.
4. Create a structure Student with name, roll, and marks.
5. Input details for an array of three Student structures.
6. Sort the students in **ascending order of marks**.
7. Display the sorted list of students clearly.

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Task 4 – (5 Marks)

Singly Linked List

Write a C++ program to:

1. Create a singly linked list to store **names**.
2. Implement **insertion** at:
    - The **beginning** of the list
    - The **end** of the list
    - A **specific position** in the list
3. Traverse and **display all names** in the list.

Task 5: Viva. (5)

-Good luck-

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